Appendix B. Evidence Engine and Recipe Ranking (Layers 3 and 4)

This appendix documents how the HepatoTwin pipeline turns the Layer 2 patient state into evidence-linked nutritional priorities (Layer 3) and ranks candidate recipes from RecipeDB against them (Layer 4, NutriTwin). It records the evidence base, the rules, the recipe criteria and the screening of the recipe corpus, and reports the evaluation set out in Section 12.2 of the proposal. All figures come from scripts and result files in the project repository.

B.1 Pipeline Overview

1. **Layer 2, patient state.** The combined model (Appendix A) gives a predicted CAP and calibrated probabilities P(CAP ≥ 248) and P(CAP ≥ 280), each with an F1-selected decision threshold.
2. **Layer 3, patient-state flags.** The probabilities and routine measurements are converted into explicit 0/1 flags using published clinical thresholds (Table B.3). Flags describe which evidence rules apply; they are not diagnoses.
3. **Layer 3, priorities.** Each evidence rule fires on a flag, sets a base priority (low, medium or high), may raise it when further flags hold, and cites rows of the evidence table (Table B.2). No priority carries a patient-level numeric target.
4. **Layer 4, recipe criteria.** Each priority is linked to one recipe-composition measure with a reference value from public dietary guidance (Table B.4). The priority level sets the weight.
5. **Layer 4, ranking.** Candidate meals are scored for health relevance, cuisine compatibility and practical feasibility (each 0–100) and combined with explicit weights (0.6, 0.2, 0.2). Recipes that meet every reference value rank first. Each recommendation reports its criterion values, reference values and supporting evidence rows.

B.2 Intervention Evidence Table

The evidence table contains 17 intervention–outcome rows from eight sources. Every citation was checked against its PubMed record. The check corrected two first authors (the British Journal of Nutrition meta-analysis is by Sangouni et al., and the Clinical Nutrition meta-analysis by Haigh et al.), completed an incomplete citation (Stern et al., 2026), and repaired two guideline rows whose fields had shifted by one column. Three rows were added to

support the saturated-fat and sugar priorities, including one null finding, so that conflicting evidence is retained.

Table B.1. Intervention evidence rows cited by the rules.

ID	Intervention	Outcome	Strength	Source
VG2015-WL5	Weight loss ≥ 5%	NASH resolution, NAS	Strong	Vilar-Gomez et al., 2015
VG2015-WL7-10	Weight loss 7–10%	NAS	Strong	Vilar-Gomez et al., 2015
VG2015-WL10-NASH	Weight loss ≥ 10%	NASH resolution	Strong	Vilar-Gomez et al., 2015
VG2015-WL10-FIB	Weight loss ≥ 10%	Fibrosis regression	Strong	Vilar-Gomez et al., 2015
SAN2021-AST-GGT	Mediterranean diet	AST, GGT	Moderate	Sangouni et al., 2022
SAN2021-ALT	Mediterranean diet	ALT (no effect)	Weak / inconclusive	Sangouni et al., 2022
STE2026-MEDDIET	Mediterranean diet	ALT, stiffness, MRI-PDFF	Moderate	Stern et al., 2026
HAI2022-DIET	Dietary interventions	ALT, AST, FLI, steatosis	Moderate	Haigh et al., 2022
HAI2022-CR	Calorie restriction	ALT, steatosis, stiffness	Moderate	Haigh et al., 2022
XIO2024-MD-LFD	Mediterranean vs low-fat	Enzymes, liver fat (no difference)	Moderate	Xiong et al., 2024
STI2023-EX	Exercise training	MRI liver fat	Moderate	Stine et al., 2023
STI2023-EXDOSE	Exercise ≥ 750 MET-min/week	MRI liver fat	Moderate	Stine et al., 2023
EASL2024-DIET	Diet quality; limit sugars, saturated fat	Liver injury	Guideline	EASL-EASD-EASO, 2024
EASL2024-PA	Physical activity	Steatosis	Guideline	EASL-EASD-EASO, 2024
LUU2018-SAT	Excess saturated vs unsaturated fat vs sugars	Liver fat	Moderate	Luukkonen et al., 2018
CHI2014-HYPER	Hypercaloric fructose	Liver fat, ALT	Moderate	Chiu et al., 2014
CHI2014-ISO	Isocaloric fructose	Liver fat, ALT (no effect)	Weak / inconclusive	Chiu et al., 2014

Strength labels follow the project scheme (STRONG, MODERATE, WEAK / INCONCLUSIVE, INSUFFICIENT), with guideline recommendations recorded separately. The Luukkonen et al. trial (n = 38) is labelled MODERATE as a controlled trial with a significant effect, although it is smaller than the scheme's "well-powered" description.

B.3 Patient State and Priority Rules

Table B.2. Thresholds used to derive patient-state flags.

Flag	Definition	Source
Steatosis likely	P(CAP ≥ 248) ≥ 0.432 (classifier threshold)	Layer 2 (Appendix A)
Moderate–severe steatosis likely	P(CAP ≥ 280) ≥ 0.365 (classifier threshold)	Layer 2 (Appendix A)
Overweight / obesity	BMI ≥ 25 / ≥ 30 kg/m²; ≥ 23 / ≥ 27.5 for Asian participants	WHO; WHO Expert Consultation, 2004

Flag	Definition	Source
Central obesity	Waist > 102 cm (men), > 88 cm (women)	NCEP ATP III
Elevated glycaemia	HbA1c ≥ 5.7%	American Diabetes Association
Dyslipidaemia	Triglycerides ≥ 150 (fasting) or ≥ 175 mg/dL (non-fasting); HDL < 40 (men) or < 50 mg/dL (women)	NCEP ATP III; EAS/EFLM, 2016
Weight management appropriate	Steatosis likely and (overweight or central obesity)	Project rule

Table B.3. Evidence rules: when each priority applies and how its level is set.

Priority	Applies when	Base level	Raised when	Evidence rows
Weight management	Weight management appropriate	Medium	High if moderate–severe steatosis or obesity	VG2015 (×3), HAI2022-CR
Dietary quality	Steatosis likely	Medium	High if moderate–severe steatosis	EASL2024-DIET, STE2026, HAI2022, SAN2021 (×2), XIO2024
Limit saturated fat	Steatosis likely	Medium	High if dyslipidaemia	EASL2024-DIET, LUU2018-SAT
Limit excess sugar	Steatosis likely	Low	Medium if elevated glycaemia or weight management appropriate	EASL2024-DIET, CHI2014-HYPER, CHI2014-ISO

The sugar priority starts low because feeding trials show harm when fructose adds surplus energy but not when it replaces other carbohydrates (CHI2014-ISO). Applied to the 3,060 held-out participants, 62% receive at least one priority, with a mean of 2.5 priorities each; participants whose steatosis is unlikely receive none.

B.4 Recipe Criteria and Candidate Screening

Table B.4. Recipe criteria linked to each priority. Scores are 100 at or beyond the reference value and fall linearly to 0 over the tolerance.

Priority	Recipe measure	Reference	Tolerance	Source of reference
Weight management	Energy per serving	≤ 600 kcal	400 kcal	Public Health England 400-600-600, 2018
Limit saturated fat	Saturated fat, % of energy	≤ 10%	10 points	WHO, 2023; DGA 2020–2025
Limit excess sugar	Total sugars, % of energy	≤ 10%	15 points	WHO sugars guideline, 2015
Dietary quality	Fibre per 1,000 kcal	≥ 14 g	14 g	DGA 2020–2025

Reference values describe the composition of a recipe and are the same for every patient; what varies by patient is which criteria apply and how they are weighted (low 1, medium 2, high 3). RecipeDB reports total rather than added sugars, so fruit and milk sugars count towards the sugar criterion.

Table B.5. Screening of the RecipeDB corpus before ranking.

Step	Rule	Effect
Nutrition plausibility	Energy > 0 and ≤ 2,500 kcal per serving; sodium ≤ 6,000 mg; macronutrient energy within 30% of stated energy	774 of 20,000 recipes (3.9%) held for review
Main meals only	Excludes non-meal categories (desserts, beverages, sauces, household and beauty items) and titles naming seasonings or condiments; energy ≥ 200 kcal per serving	10,108 candidate meals remain
Dietary preference	Every diet tag must be allowed by the requested diet	About 1,000 recipes have conflicting tags (e.g. a prawn dish tagged vegan)

B.5 Evaluation of the Ranking Engine

Following Section 12.2 of the proposal, the ranking was evaluated on 300 randomly selected held-out participants who had at least one priority (of 1,902 eligible). For each participant, the ten highest-ranked recipes from each method were checked against the three criteria: health (the recipe meets the reference value of every criterion that applies to the patient), cuisine (it matches the requested cuisine) and feasibility (total time ≤ 60 minutes, ≤ 15 ingredients and ≤ 12 steps). NHANES does not record cuisine preferences, so each participant was assigned one at random from the five cuisines with at least 100 candidate meals. Random selection was repeated 20 times per participant.

Table B.6. Share of each top-10 list meeting the criteria (mean over 300 participants).

Method	Health	Cuisine	Feasibility	All three
Multi-criteria, health-gated (selected)	1.000	1.000	0.899	0.899
Multi-criteria, weighted only	0.749	1.000	0.982	0.749
Health score only	1.000	0.057	0.400	0.017
Nutrition only (non-personalised)	1.000	0.057	0.400	0.017
Random selection	0.057	0.053	0.596	0.003

The multi-criteria ranking is the only method that satisfies all three criteria for most recommendations. Ranking on health alone finds healthy recipes but ignores cuisine and practicality, and random selection rarely meets the health criteria. Without the health gate, the weighted score traded health for cuisine in a quarter of recommendations; with it, every recommendation met the evidence-linked criteria and 90% met all three. The gate was therefore adopted as the default. The health-only and nutrition-only baselines coincide because 565 candidates meet every reference value, so both select among the same tied recipes.

The sensitivity analysis reran the selected ranking over every combination of weights in steps of 0.1 (Figure B.1). The top ten was identical to the default whenever cuisine had a weight of at least 0.2, and the mean overlap across the whole grid was 0.90. Rankings changed only when the cuisine weight fell to 0.1 and feasibility dominated. Because of the gate, the health weight has little effect on which recipes are recommended.

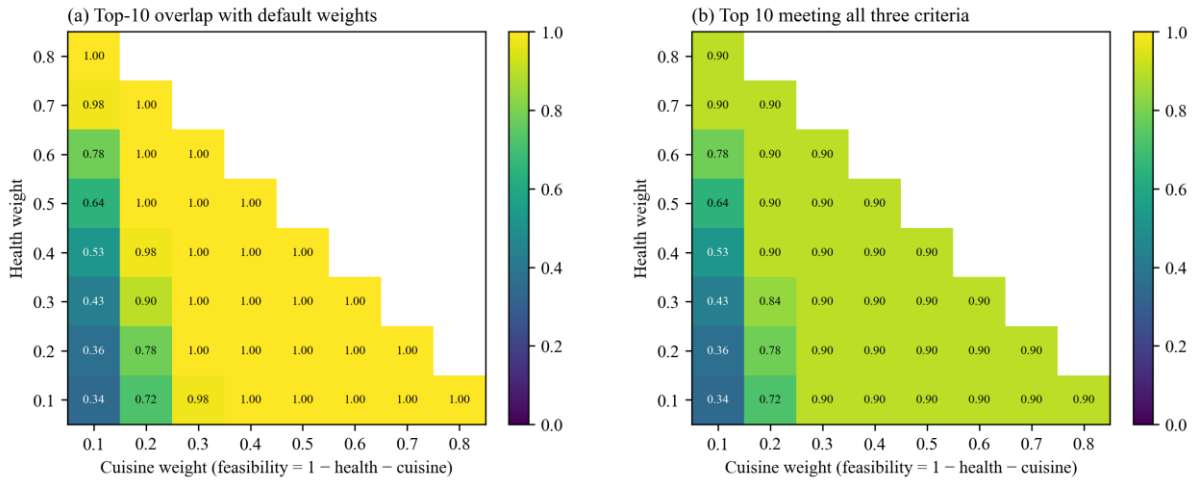


Figure B.1. Weight sensitivity of the selected ranking over 300 held-out participants. (a) Mean share of the default top ten that is retained. (b) Mean share of the top ten meeting all three criteria.

B.6 Worked Example

Participant 109349 (held out from training; measured CAP 338 dB/m) has calibrated probabilities of 0.96 for CAP ≥ 248 and 0.78 for CAP ≥ 280. Layer 3 assigns high priority to weight management, dietary quality and limiting saturated fat (the last because of dyslipidaemia), and medium priority to limiting excess sugar. With Asian cuisine requested, 9,844 candidate meals are scored, 565 meet every reference value and 264 are held for data review.

Table B.7. Top three recommendations for participant 109349 (Asian cuisine requested).

Rank	Recipe	Energy (kcal)	Sat. fat (% energy)	Sugars (% energy)	Fibre (g/1,000 kcal)	Overall
1	Avocado and Prawns in Wasabi	234	8.8	1.4	29.9	96.3
2	Hummus Without Tahini	335	7.3	0.6	20.3	95.5
3	Creamy Chickpea Salad	207	7.4	3.7	20.3	95.4

B.7 Limitations

- Reference values come from population dietary guidance, not from liver-specific dose–response evidence; the evidence rows justify the direction of each criterion, not its threshold.
- Criterion scores reach their ceiling once a recipe meets the reference, so the engine does not distinguish between compliant recipes on health grounds; cuisine and feasibility order them.
- Most participants with priorities receive all four, so personalisation comes mainly from which priorities apply (for example, no weight-management criterion for lean participants) and from the patient's preferences, rather than from the priority levels.
- Cuisine preferences were simulated, and 73% of RecipeDB recipes have the cuisine label "Other", which limits cuisine matching. Cuisine compatibility uses exact label matches rather than the ingredient co-occurrence measure described in the proposal.
- RecipeDB nutrition is per serving as reported by recipe authors; implausible values are screened out but plausible errors remain. Total sugars stand in for added sugars.
- The evaluation measures whether recommendations satisfy the defined criteria. It does not show that following them changes liver fat; the planned pilot evaluation of recipe appropriateness has not yet been run.